Sebastian De Leon

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Summary

Computer engineer with hands-on experience in full-stack development, embedded systems, and game development, focused on building reliable software and hardware-integrated solutions that solve real-world problems. Strong foundation in software engineering, systems thinking, rapid prototyping, and cross-functional collaboration, with experience debugging complex issues and delivering scalable products. Passionate about sustainable energy, intelligent systems, and high-impact engineering that advances performance, safety, and innovation at Tesla.

Education

Western Michigan University — Sep 2025 – Dec 2027
B.S. in Computer Engineering, GPA: 3.89 — Kalamazoo, MI

Instituto Tecnológico de Santo Domingo — Nov 2022 – Jul 2025
B.E. in Computer Science (transferred), GPA: 3.89 — Santo Domingo, Dom. Rep.

Experience

Junior Software Engineer — Jul 2024 – Jul 2026
Reispsolutions S.R.L. — Santo Domingo, Dom. Rep.

- Architected a distributed Learning Management System (LMS) for a multi-brand restaurant corporation in the Dominican Republic, scaling training infrastructure across dozens of branch locations using ASP.NET.
- Built production-oriented ASP.NET Core services with .NET 9, CQRS, DDD, and Clean Architecture, implementing controllers, application services, domain models, and EF Core persistence across multiple backend services.
- Developed an end-to-end clinical management web application that automated patient intake flows, integrated calendar scheduling, and deployed automated SMS/email reminders for upcoming appointments.
- Worked on self-service Android kiosk applications for large-format tablets across multiple restaurant brands, leveraging Jetpack Compose to deliver a highly responsive, modern, and intuitive in-store ordering experience.

Manufacturing & Telemetry Assistant — Aug 2024 – Apr 2025
Instituto Tecnológico de Santo Domingo's NASA HERC College Team — Santo Domingo, Dom. Rep.

- Designed and implemented Task A for the 2025 NASA HERC rover: a lead-screw actuator driven by a stepper motor that moved a sensor vertically to detect whether a container holding liquid contained water.
- Built the rover control system around an ESP32-S3, reading PWM signals from a commercial RC receiver and implementing a state machine to map controller inputs to rover motion and task execution.
- Contributed to a rover platform that traversed simulated Mars and Moon terrain while performing additional mission tasks such as distance measurement and object detection under competition constraints.
- Led the full-stack architecture for the official Apolo 27 website, including frontend and backend development.
- Managed the development of proprietary games and digital tools for STEM outreach across the Dominican Republic.

Projects

Velo: Movie Matchmaker — Sep 2025 – Present
Shipped mobile app — React Native, Supabase

- Built a cross-platform Expo/React Native movie matchmaking app where groups create or join sessions, swipe on shared movie decks, and receive ranked watchlists using weighted voting, veto logic, and third-party API metadata.
- Designed a Clean Architecture codebase with TypeScript, Zustand, Supabase Auth/Postgres/Realtime/Storage, repository abstractions, dependency injection, and Edge Functions that securely proxy and cache third-party API data.
- Implemented production-focused features including solo discovery, QR/code-based session joining, persistent swipe progress, guest-to-account data migration, Letterboxd imports, taste-profile recommendations, five-language localization, and RevenueCat/AdMob monetization.

Quadruped Robot Dog — May 2024 – Aug 2024
Custom mobile robotics prototype — FreeCAD, Arduino, ESP32-S3, PCB Design

- Designed and fabricated a custom quadruped robot using FreeCAD; selected electromechanical components and prototyped the embedded control circuitry on a perfboard.
- Developed a Wi-Fi-enabled web interface hosted on an ESP32-S3 Xiao Seed, allowing real-time remote smartphone control of the robot's walking and rotation behaviors.
- Programmed sequential gait poses and motion primitives via a servo driver to enable stable locomotion and turning within a compact embedded platform.

Technical Skills

- **Programming Languages**: C, C++, SQL, JavaScript, TypeScript
- **Frameworks & Tools**: Git, React, React Native, .NET, Linux, ESP-IDF, Docker
- **Software & Tools**: OrCAD X, KiCad, LTspice, SolidWorks, Fusion 360
- **Languages**: Spanish (native), English (Fluent)

Achievements & Certifications

- WMU's Computer Club annual hackathon winner (2025)
- Harvard's CS50x: CS50's Introduction to Computer Science (edX)
- WMU College of Engineering and Applied Sciences Dean's List (all semesters)